

COMPREHENSIVE ONCOLOGY CONSULTATION REPORT

Patient Submitter ID: TCGA-AA-3666 | Tumor: Colon Adenocarcinoma (COAD) | Stage: Stage IIA

|                 |                   |                    |                                  |
|-----------------|-------------------|--------------------|----------------------------------|
| TCGA Project:   | TCGA-COAD         | cBioPortal Study:  | coadread_tcga_pan_can_atlas_2018 |
| Gender / Age:   | male / 59 yrs     | Vital Status / OS: | Alive (52.1 mo)                  |
| Hybrid ML Risk: | 58.2% (HIGH RISK) | VQC / Classical:   | 54.1% / 61.0%                    |

FOUNDATIONAL 6-DATABASE MULTI-OMICS MUTATION PROFILE (GDC, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl)

| Gene | Locus (GRCh38) | Transcript      | Protein Change                  | ClinVar Status                                          | COSMIC ID                               | gnomAD Pop AF | VAF % |
|------|----------------|-----------------|---------------------------------|---------------------------------------------------------|-----------------------------------------|---------------|-------|
| APC  | chr5:5q22.2    | ENST00000257430 | p.R1450*<br>Nonsense Truncation | Pathogenic (FAP / Sporadic CRC)<br>VCV000000898         | COSV53492104<br>FATHMM: 0.99 Pathogenic | 0.000012      | 47.3% |
| KRAS | chr12:12p12.1  | ENST00000311936 | p.G12D<br>Activating Transition | Oncogenic Driver (Anti-EGFR Resistance)<br>VCV000012583 | COSV55546255<br>FATHMM: 0.99 Pathogenic | 0.000008      | 36.9% |

QUANTUM VARIATIONAL ROTATION STATE ANGLES (θ\_VQC / R\_Y / CNOT)

| Gene Symbol | Canonical Ensembl ID | VAF Hotspot Frequency | Variational Angle θ (Radians) | Qubit Hilbert Space Projection                               |
|-------------|----------------------|-----------------------|-------------------------------|--------------------------------------------------------------|
| APC         | ENSG00000134982      | 47.3%                 | θ = 2.1900 rad (125.5°)       | $ \psi\rangle = \cos(1.095) 0\rangle + \sin(1.095) 1\rangle$ |
| KRAS        | ENSG00000133703      | 36.9%                 | θ = 2.2150 rad (126.9°)       | $ \psi\rangle = \cos(1.107) 0\rangle + \sin(1.107) 1\rangle$ |

PRECISION ONCOLOGY CLINICAL ACTION & PROTOCOL RECOMMENDATION

Recommended Therapeutic Pathway:

Complete mesocolic excision staging review, MSI/MMR status confirmation for immunotherapy eligibility.

Genomic Biomarker Summary: Identified somatic hotspot mutations in APC (p.R1450\*), KRAS (p.G12D). Confirmed pathological classification across ClinVar and COSMIC registries with extreme somatic rarity in gnomAD exomes.

NOTICE & RESEARCH DISCLAIMER: This report is generated by the Blazeфинix Quantum-Classical Multi-Omics Research Platform using open-access academic cancer genomics datasets (NCI GDC TCGA, ClinVar, COSMIC, cBioPortal, gnomAD, Ensembl). It is calibrated strictly for computational research, translational bioinformatics, and clinical decision support benchmarking, not for standalone autonomous medical diagnosis.